THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

Dec 18, 2021 TIME: 9:30 AM – 12:30 PM

I. Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

II. Atomic weights: H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

III. Properties of water:
Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg·°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

IV. Ideal gas law constant = 0.082056 L·atm/(K·mol)
Henry's Law coefficient for O_2 (25 °C) = 0.001263 mol/(L·atm)
Henry's Law coefficient for CO_2 (25 °C) = 0.033363 mol/(L·atm)
Dissociation constant of water (25 °C) $K_w = 1\times10^{-14}$
Dissociation constants for carbonate system: $K_1 = 4.47\times10^{-7}$, $K_2 = 4.68\times10^{-11}$

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (10 multiple choice questions with 3 marks each)

1. Which of the following units does NOT describe mass concentration for the liquid? (a) mg/L, (b) ppb, (c) M, or (d) g/m^3.
2. What is the volume of 1 mole ideal gas law at 1 atm and 25 °C? (a) 22.4 L, (b) 24.5 L, (c) 22.0 L, or (d) 22.8 L.
3. What is the amount of oxygen required to fully oxidize 1.67 mM glucose (chemical formula $C_6H_{12}O_6$)? (a) 280 mg/L, (b) 320 mg/L, (c) 350 mg/L, or (d) 300 mg/L.
4. Suppose the gas above the soda in a bottle of soft drink is pure CO_2 at a pressure of 2 atm. What is mass concentration of CO_2 at 25 °C? (a) 0.1 mol/L, (b) 1.5 g/L, (c) 0.067 mol/L, or (d) 2.9 g/L.
5. Which of the following pollutants is NOT oxygen demanding organic waste? (a) phenol, (b) protein, (c) nitrate, or (d) pesticide.
6. Which of the following does NOT describe COD test? (a) it is not suitable for samples containing toxic substances, (b) it can be completed within 3 hours, (c) it can not provide information of biodegradation rate, or (d) it uses strong oxidants.
7. Which of the following chemicals is NOT a commonly used coagulant? (a) $FeCl_3$, (b) polyelectrolytes, (c) $Ca(OH)_2$, or (d) $Al_2(SO_4)_3 \cdot 14H_2O$.
8. Which of the following is NOT found in a prokaryotic cell? (a) cell membrane, (b) nucleus, (c) cytoplasm, (d) ribosome, or (e) cell wall.
9. Which of the following is NOT a characteristic of wastewater? (a) fecal coliform, (b) SOR, (c) Ammonia-nitrogen, (d) TSS, or (e) BOD.
10. Which of the following is NOT a typical greenhouse gas? (a) methane, (b) ozone, (c) nitrous oxide, (d) hydrofluorocarbons, or (e) sulfur dioxide.

Section B. (Three questions with unequal marks)

11. (a) Briefly explain the following terminologies (use < 80 words for each term).
    i. NBOD
    ii. Eutrophication
    iii. Superficial velocities
    iv. Stable air

(12 marks)

(b) A wastewater treatment plant discharges 6×10^4 L/min of effluent having an ultimate BOD of 40.0 mg/L into a stream flowing at 10.0 m^3/s. Just upstream from the discharge point, the stream has an ultimate BOD of 3.0 mg/L. The deoxygenation constant k_d is estimated at 0.22/day.

i. Assuming complete and instantaneous mixing, find the ultimate BOD of the mixture of waste and river just downstream from the outfall.
ii. Assuming a constant cross-sectional area for the stream equal to 55 m^2, what ultimate BOD would you expect to find at point 10,000 m downstream?
iii. The wastewater has a DO equal to 4.0 mg/L when it is discharged. The river has its own DO, just upstream from the outfall, equal to 8.0 mg/L. Find the initial oxygen deficit of the mixture just downstream from the discharge point. The temperature of wastewater and river are both 15 °C. (Hints: The saturated DO in the river at 15 °C is 10.08 mg/L)

(12 marks)

12. (a) 0.5 g CaSO4(s) is added to 1.0 L of pure water. Immediately, the solid begins to dissolve according to the following reaction: $CaSO_4 \leftrightarrow Ca^{2+} + SO_4^{2-}$. What is the concentration of dissolved sulfate in the water once equilibrium is achieved. (Hint: Solubility product constant, $K_{sp,CaSO4} = 2\times10^{-5}$).

(6 marks)

(b) A solution is initially supersaturated with Ca^{2+} and CO_3^{2-} ions by rapid addition of $CaCO_3$, such that the concentrations of each are 2.0×10^{-3} mol/L. The water's pH is held at 8.5. When equilibrium is finally reached, determine the concentration of free Ca^{2+} in the water and the mass of $CaCO_3(s)$ per liter of water.
(Hints: Assuming $[Ca^{2+}] = [H_2CO_3]+[HCO_3^-]+[CO_3^{2-}]$)

(12 marks)

(c) Briefly discuss the effect of global warming on ocean acidification (**< 300 words**).

(8 marks)

13. As an engineer working with the city's drainage services department, you have been tasked to address a river pollution problem caused by illegal discharge of municipal wastewater. Figure 1 shows the flow diagram of a proposed municipal wastewater treatment system. Answer the following questions:

(a) If the untreated municipal wastewater is discharged to the river, explain what will happen to the dissolved oxygen (DO) and the ecosystem in the river.

(7 marks)

(b) Explain the function of the components in the flow diagram design (Figure 1).

(7 marks)

(c) For the sludge processing unit, your subordinate has suggested two possible options:
- centrifugation
- anaerobic digestion

Explain the differences in sludge disposal method between the two options and how this would affect the treated sludge quality.

(6 marks)

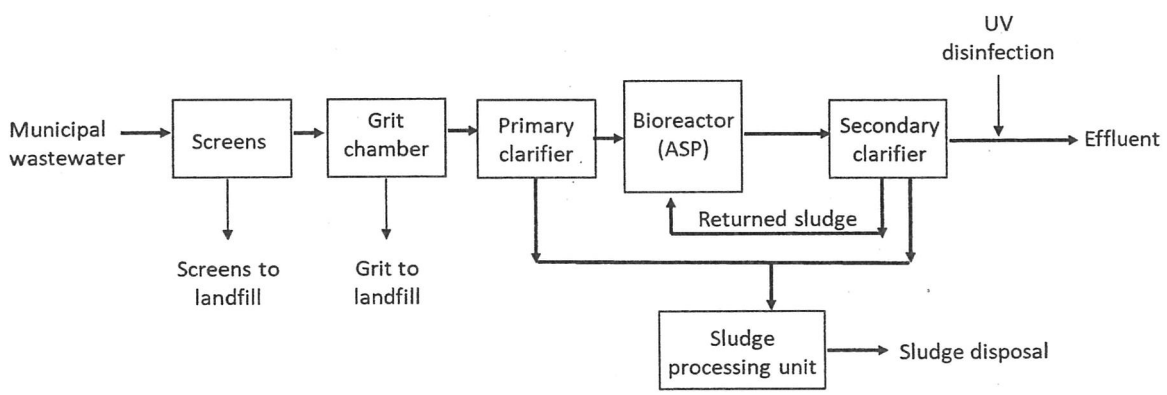


Figure 1. Flow diagram of a proposed municipal wastewater treatment system.

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